

Memo: which lines to test in 2008

To: Head of the maize pipeline (115-day, first-stage testcrosses) · **From:** Team ProMaize · **January 2008**

The decision

We have plots for about 30% of the 15,962 new lines (4,790 lines at about seven locations each). A line we do not test this year can never be advanced. None of the 157 families behind these lines has been in a field before.

What we recommend

Test the same share of every family, and let the DNA choose which brothers and sisters get the plots. The list is ready: docs/advance_2008.csv, 4,789 lines, every family represented.

Why

1. **Every family is new, so a family's average can only be guessed from its parents.** That guess is good when the parents have been tested before and weak when they have not.
2. **This year the parents are mostly unknown.** 24 of the 157 families have neither parent tested before, and 88 more have only one. Two earlier years looked like this: 2003 and 2007.
3. **In those years, spreading the plots evenly cost little:** \$2.02 and 33 cents an acre compared with betting on the best-looking families, against \$4.50 to \$7.50 in years with well-known parents, and it kept about twice as many families alive. We knew that in January. (Checked afterwards: in 2008 it cost 8 cents an acre, again for twice the families.)
4. **Inside a family, the DNA ranking works every year.** It reliably tells which siblings are better, even when it cannot tell which family is better. The even split uses exactly that part.

The two options

	Conservative (recommended)	Aggressive
What it does	same share of plots to every family, DNA picks the siblings	every line ranked by predicted \$/acre
Families kept (2008 lines)	157, about 138 "effective"	132, about 71 "effective"
Value in the 2008 field	\$5.0 per acre over random picks	\$5.1 per acre over random picks
Average over six past seasons	\$5.5 per acre	\$8.8 per acre
Best when	the parents are new (2003, 2007, 2008)	the parents have a long record (2004-2006)
Risk	leaves a little value when the family guess is good	concentrates plots on guesses we cannot yet trust

In 2008 both options clearly beat the standard genomic method, which realised \$1.2 per acre; over six seasons the aggressive option averaged \$8.8 against its \$5.5. With the standard method we would have needed 1,277 more lines in 2008 (about 8,900 plots) to find as many of the season's real winners.

How far to trust this

- We checked it the only fair way: predict each past season from the seasons before it, then compare with what the field did. Six seasons, 2003 to 2008; it beat the standard method in all six.
- It is a first screen, not a final answer: field plots are noisy (a line's yield swings more from field to field than lines differ from each other), so no prediction can be close to perfect.
- Every line comes with a range it will probably land in. In 2008, 90% of lines landed in theirs.
- We predict the line's value across all locations, not location by location: we tested location by location and it could not be predicted, because each line is tested in one year only.

What we need from you

1. The real plot budget, if it is not 30%.
2. Your drying cost, moisture target and harvest loss from lodging, so the dollars are your dollars.
3. Confirmation that at this stage lines are judged on how good a parent they are (their effect across testers), not on the one tester they were crossed to.

The interactive version, with every number and both options, is at promaize.stipe.app.